

```
1 // Problem 1
2 class Book {
3     Book(this.title, this.author, this.price);
4
5     String title;
6     String author;
7     double price;
8
9     void discountedPrice(double discountPercentage) {
10         double discountAmount = price * (discountPercentage / 100);
11         double finalPrice = price - discountAmount;
12         print(
13             'The discounted price of "$title" by $author is:
14             \${finalPrice.toStringAsFixed(2)}',
15         );
16     }
17
18 // Problem 2
19 // Base class
20 class Employee {
21     Employee(this.name, this.salary);
22
23     String name;
24     double salary;
25 }
26
27 // Subclass Manager
28 class Manager extends Employee {
29     Manager(super.name, super.salary, this.department);
30
31     String department;
32
33     void displayDetails() {
34         print(
35             '\nManager Name: $name, Salary: \${salary.toStringAsFixed(2)},
36             Department: $department',
37         );
38     }
39
40 // Subclass Developer
41 class Developer extends Employee {
42     Developer(super.name, super.salary, this.programmingLanguage);
43
44     String programmingLanguage;
45
46     void displayDetails() {
47         print(
48             'Developer Name: $name, Salary: \${salary.toStringAsFixed(2)},
49             Programming Language: $programmingLanguage',
50         );
51     }
52
53 // Problem 3
54 // Abstract class Applicant
55 abstract class Applicant {
56     void turnOn();
57     void turnOff();
```

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58 }
59
60 // Subclass Fan
61 class Fan extends Applicant {
62   @override
63   void turnOn() {
64     print("\nFan is now running");
65   }
66
67   @override
68   void turnOff() {
69     print("Fan is now stopped");
70   }
71 }
72
73 // Subclass Light
74 class Light extends Applicant {
75   @override
76   void turnOn() {
77     print("\nLight is switched ON");
78   }
79
80   @override
81   void turnOff() {
82     print("Light is switched OFF");
83   }
84 }
85
86 void main() {
87   // Testing Problem 1
88   Book book1 = Book("The Alchemist", "Paulo Coelho", 15.99);
89   Book book2 = Book("1984", "George Orwell", 12.49);
90
91   book1.discountedPrice(10);
92   book2.discountedPrice(20);
93
94   // Testing Problem 2
95   Manager manager = Manager("Alice", 75000, "Sales");
96   Developer developer = Developer("Bob", 65000, "Dart");
97
98   manager.displayDetails();
99   developer.displayDetails();
100
101   // Testing Problem 3
102   Fan fan = Fan();
103   Light light = Light();
104
105   fan.turnOn();
106   fan.turnOff();
107   light.turnOn();
108   light.turnOff();
109 }
110
```